

TASK 1

Task 1.1: Initialize a new Git repository for a simple "Hello World" Python Flask application (provided to you). Add the initial code and commit it to the main branch.

```
C:\Windows\System32\cmd.e X + v
Microsoft Windows [Version 10.0.26200.8457]
(c) Microsoft Corporation. All rights reserved.

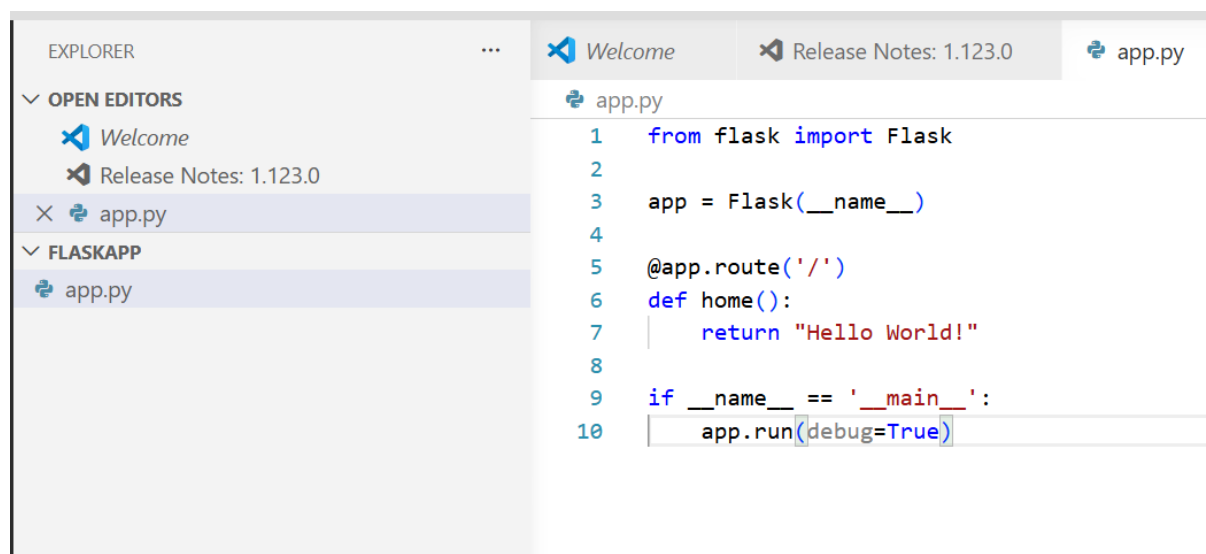
C:\Desktop\SYMBI\SEM 7\DevOps-Lab>mkdir FlaskApp

C:\Desktop\SYMBI\SEM 7\DevOps-Lab>cd FlaskApp

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\FlaskApp>
```

```
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\FlaskApp>code .

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\FlaskApp>
```



```
1 from flask import Flask
2
3 app = Flask(__name__)
4
5 @app.route('/')
6 def home():
7     return "Hello World!"
8
9 if __name__ == '__main__':
10     app.run(debug=True)
```

PROBLEMS TERMINAL ...

 Code + -   ... |  X

```
PS C:\Desktop\SYMBI\SEM 7\DevOps-Lab\FlaskApp> python -u "c:\Desktop\SYMBI\SEM 7\DevOps-Lab\FlaskApp\app.py"
```

```
File "c:\Desktop\SYMBI\SEM 7\DevOps-Lab\FlaskApp\app.py", line 1, in
<module>
```

```
    from flask import Flask
```

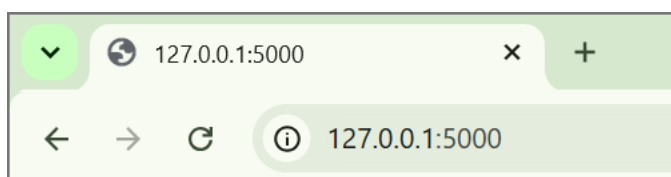
```
ModuleNotFoundError: No module named 'flask'
```

```
● PS C:\Desktop\SYMBI\SEM 7\DevOps-Lab\FlaskApp> python --version
Python 3.12.4
```

```
○ PS C:\Desktop\SYMBI\SEM 7\DevOps-Lab\FlaskApp> █
```

```
○ PS C:\Desktop\SYMBI\SEM 7\DevOps-Lab\FlaskApp> pip install flask
WARNING: Retrying (Retry(total=4, connect=None, read=None, redirect=None, status=None))
after connection broken by 'ReadTimeoutError("HTTPConnectionPool(host='pypi.org', port=
443): Read timed out. (read timeout=15)")': /simple/flask/
Collecting flask
  Downloading flask-3.1.3-py3-none-any.whl.metadata (3.2 kB)
Collecting blinker>=1.9.0 (from flask)
  Downloading blinker-1.9.0-py3-none-any.whl.metadata (1.6 kB)
Collecting click>=8.1.3 (from flask)
  Downloading click-8.4.2-py3-none-any.whl.metadata (2.6 kB)
Collecting itsdangerous>=2.2.0 (from flask)
  Downloading itsdangerous-2.2.0-py3-none-any.whl.metadata (1.9 kB)
Collecting jinja2>=3.1.2 (from flask)
  Using cached jinja2-3.1.6-py3-none-any.whl.metadata (2.9 kB)
Collecting markupsafe>=2.1.1 (from flask)
  Using cached markupsafe-3.0.3-cp312-cp312-win_amd64.whl.metadata (2.8 kB)
```

```
○ PS C:\Desktop\SYMBI\SEM 7\DevOps-Lab\FlaskApp> python app.py
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a
roduction WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 759-863-445
█
```



Hello World!

```
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\FlaskApp>git init
Initialized empty Git repository in C:/Desktop/SYMBI/SEM 7/DevOps-Lab/FlaskApp/.git/

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\FlaskApp>git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    app.py

nothing added to commit but untracked files present (use "git add" to track)
```

```
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\FlaskApp>git add .

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\FlaskApp>git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   app.py
```

```
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\FlaskApp>git commit -m "Initial commit with Hello World Flask app"
[master (root-commit) ad3fbd4] Initial commit with Hello World Flask app
1 file changed, 10 insertions(+)
create mode 100644 app.py
```

 New repository



Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).

Required fields are marked with an asterisk ().*

1

General

Owner *

 yashpatelSIT ▾

Repository name *

FlaskApp

✓ FlaskApp is available.

Great repository names are short and memorable. How about [curly-tribble](#)?

Description

0 / 350 characters

2

Configuration

Choose visibility *

Choose who can see and commit to this repository

 Public ▾

2 Configuration

Choose visibility *

Choose who can see and commit to this repository

 Public ▾

Add README

READMEs can be used as longer descriptions. [About READMEs](#)

Off ☐

Add .gitignore

.gitignore tells git which files not to track. [About ignoring files](#)

No .gitignore ▾

Add license

Licenses explain how others can use your code. [About licenses](#)

No license ▾

Create repository

```
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\FlaskApp>git remote -v
origin https://github.com/yashpatelSIT/FlaskApp.git (fetch)
origin https://github.com/yashpatelSIT/FlaskApp.git (push)
```



yashpatelSIT Initial commit with Hello World Flask app

ad3fbd4 · 2 days ago



app.py

Initial commit with Hello Worl...

2 days ago

Task 1.2: Create a new branch named feature/user-auth. Make a small modification (e.g., add a new print statement) in this branch, commit the changes, and push the branch to a remote repository (e.g., GitHub/GitLab).

```
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\FlaskApp>git checkout -b feature/user-auth
Switched to a new branch 'feature/user-auth'
```

```
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\FlaskApp>git branch
* feature/user-auth
  main
```

```
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\FlaskApp>git commit -m "Added user authentication feature"
On branch feature/user-auth
nothing to commit, working tree clean

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\FlaskApp>git push -u origin feature/user-auth
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 354 bytes | 177.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature/user-auth' on GitHub by visiting:
remote:   https://github.com/yashpatelSIT/FlaskApp/pull/new/feature/user-auth
remote:
To https://github.com/yashpatelSIT/FlaskApp.git
 * [new branch]      feature/user-auth -> feature/user-auth
branch 'feature/user-auth' set up to track 'origin/feature/user-auth'.

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\FlaskApp>
```

Task 1.3: Simulate a conflict: On the main branch, modify the *same line* you changed in feature/user-auth. Commit this change. Now, attempt to merge feature/user-auth into main. Resolve the merge conflict manually and commit the resolution. Push the updated main branch. (1.5 Marks)

```
PS C:\Desktop\SYMBI\SEM 7\DevOps-Lab\FlaskApp> git log
--oneline
1d56973 (HEAD -> main, origin/main) Resolved merge conflict
ce94bea (origin/feature/user-auth, feature/user-auth) Added user authentication feature
c30ca1d Updated welcome message on main
eea5da1 Update app.py
163d699 Added user authentication feature
ad3fbd4 Initial commit with Hello World Flask app
PS C:\Desktop\SYMBI\SEM 7\DevOps-Lab\FlaskApp>
```

```
Welcome
app.py
app.py
1  from flask import Flask
2
3  app = Flask(__name__)
4
5  print("Welcome to the Flask Application")
6  print("User Authentication Feature")
7
8  @app.route('/')
9  def home():
10     return "Hello World!"
11
12  if __name__ == '__main__':
13     app.run(debug=True)
```

github.com/yashpatelSIT/FlaskApp

Ask Google

yashpatelSIT / FlaskApp

Code Issues Pull requests Actions Projects Wiki Security and quality Insights Settings

FlaskApp Public

Pin Watch 0 Fork 0 Starred 1

main 2 Branches 0 Tags

Go to file

Add file Code

yashpatelSIT Resolved merge conflict 1d56973 · yesterday 6 Commits

app.py Updated welcome message on main yesterday

README

Add a README

Help people interested in this repository understand your project.

Add a README

About

No description, website, or topics provided.

Activity

1 star

0 watching

0 forks

Releases

No releases published

Create a new release

Packages



main 2 Branches 0 Tags

Switch branches/tags



Find or create a branch...

Branches

Tags

✓ main

default

feature/user-auth

View all branches

Branches

- Overview
- Yours
- Active
- Stale
- All

 Search branches...

Default

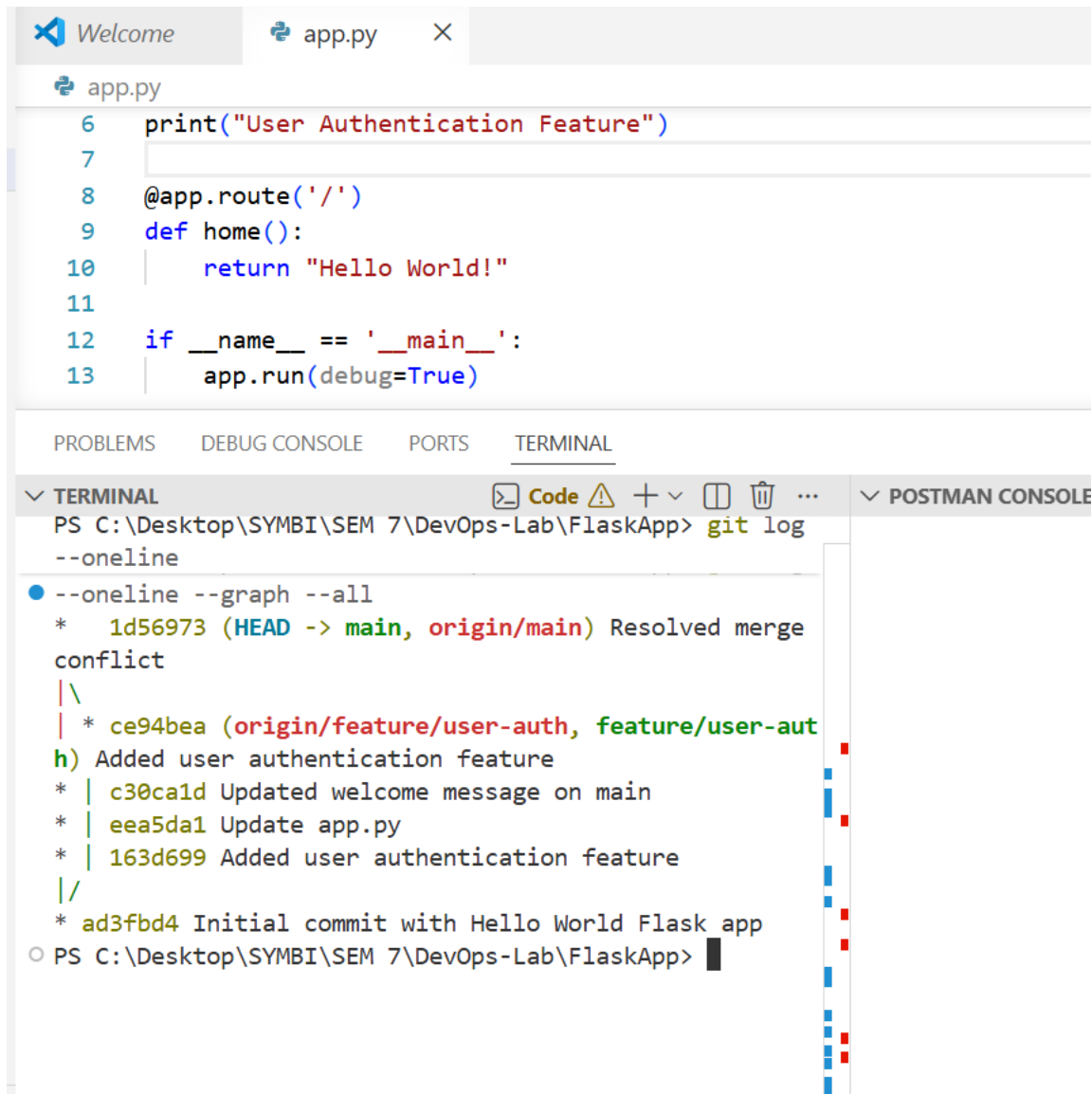
Branch	Updated
main	 8 minutes ago

Your branches

Branch	Updated
feature/user-auth	 yesterday

Active branches

Branch	Updated
feature/user-auth	 yesterday



The screenshot shows the Visual Studio Code interface. The top bar has tabs for 'Welcome' and 'app.py'. The editor window displays the code for 'app.py' with line numbers 6 through 13. The code is a simple Flask application with a home route. Below the editor, the 'TERMINAL' panel is active, showing the output of a 'git log --oneline' command. The log shows a series of commits, including an initial commit and several updates, with a merge conflict resolved.

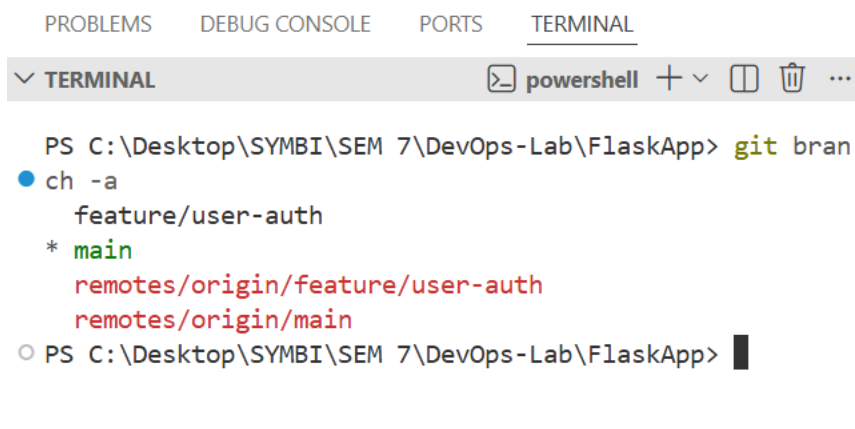
```
6 print("User Authentication Feature")
7
8 @app.route('/')
9 def home():
10     return "Hello World!"
11
12 if __name__ == '__main__':
13     app.run(debug=True)
```

PROBLEMS DEBUG CONSOLE PORTS TERMINAL

✓ TERMINAL Code + - [] [] ... POSTMAN CONSOLE

PS C:\Desktop\SYMBI\SEM 7\DevOps-Lab\FlaskApp> git log --oneline

- --oneline --graph --all
- * 1d56973 (HEAD -> main, origin/main) Resolved merge conflict
- | \
- | * ce94bea (origin/feature/user-auth, feature/user-auth) Added user authentication feature
- * | c30ca1d Updated welcome message on main
- * | eea5da1 Update app.py
- * | 163d699 Added user authentication feature
- | /
- * ad3fbd4 Initial commit with Hello World Flask app
- PS C:\Desktop\SYMBI\SEM 7\DevOps-Lab\FlaskApp>



The screenshot shows the Visual Studio Code interface with the 'TERMINAL' panel active. The terminal shows the output of a 'git branch' command, creating a new branch named 'feature/user-auth' based on the current branch 'main'.

PROBLEMS DEBUG CONSOLE PORTS TERMINAL

✓ TERMINAL powershell + - [] [] ...

PS C:\Desktop\SYMBI\SEM 7\DevOps-Lab\FlaskApp> git branch

- ch -a
- feature/user-auth
- * main
- remotes/origin/feature/user-auth
- remotes/origin/main
- PS C:\Desktop\SYMBI\SEM 7\DevOps-Lab\FlaskApp>